Abhishek Singh

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Research Interests

Machine Learning, Computer Vision, Deep Learning, Robotics

Academic History

Under Graduate	Senior Secondary School	Secondary School Certificate
Indian Institute of Technology,Varanasi	D.A.V Public School, Kota	St. Peter's School, Panchgani
B.Tech in Electrical Engineering Expected 2019	Central Board of Secondary Education (C.B.S.E),(2015)	Indian Certificate of Secondary Education(I.C.S.E),(2013)
Cumulative Grade Point Average-9.09/10	Percentage-91.5%	Percentage-92.2%
Project-Localization of Mobile robots	Computer Science-98/100	Mathematics-97/100

Skills and Certifications

Programming/ Scripting Languages- C, C++, Python, Java, CSS, PHP, HTML
Frameworks and Tools- Tensorflow, Keras, CUDA, Git, Grafana, Prometheus, Data Structure and Algorithms
MOOCs- Deep Learning Specialization, Sequence Models, CNNs, Hyper Parameter Tuning, Optimization, Structuring Machine Learning Projects, Neural Network and Deep Learning

Research, Teaching and Internship Experience

Research Internship

Design and Innovation Hub at IIT(BHU) Varanasi **May –July 2017**

- Solar Energy Forecasting using Tensorflow, applied LSTM and Regression, accuracy 16% MAPE.
- Applied Random Forest, LightGBM and Ridge Regression on the same dataset to compare accuracy.

Design and Innovation Hub at IIT(BHU) Varanasi **May-July 2016**

- Smart Metering System, using GSM and Arduino, created a website for Electricity Users.
- Features- Online Monitoring, Electricity Cut-off, Monitoring through a SMS.

Teaching Assistant

1. Basic Electrical Engineering (UG Sophomore Level Course) July- December 2018
2. Computer Programming and Linux Concepts (UG freshman Level Course) January-May 2018

Software Engineering Internship at Indus Valley Partners, Noida, India

- Created a Cloud watch Using Prometheus, Grafana and Alert manager to Send Alerts, Monitor Servers on Grafana Dash Boards. https://github.com/asking28/resource_monitoring/wiki
- Features- Sending Email alerts of Server Usage, Monitoring through Prometheus and grafana dashboards.

Honors, Awards and Presentations

1. Poster Presentation at Institute Day IIT (BHU). 2017
2. Project Presentation at IIT Guwahati and secured top-5 positions among 85 projects. 2016
3. Qualified KVPY ,Research Scholarship by Indian Government (Top 99.4 percentile) 2015
4. Cleared IIT-JEE Advanced with top 99.99 Percentile. 2015

Projects

Mobile Robot Localization

- Applied Kalman Filtering on for positioning using Accelerometer and four Ultrasound sensors.

Object Recognition and Classification

- Object detection using YOLOv2 and YOLOv3 and YOLOv3_tiny architectures using Python and Tensorflow. Trained on Custom dataset and increased data using Tensorflow data Augmentation. [Project]
- Evaluated Recall metrics of above architectures, with YOLOv3 being highest at 70% on testing images.

Electricity Price Forecasting

- Price forecasting with Neural Networks on MATLAB, MAPE of 6.5%, and MAE of 5.2%.
- Analyzed hourly and weekly demands of Electricity Load and their corresponding prediction errors.

Extracurricular Activities

1. Won Gold Medal in Javelin Event in Spardha Sports Meet 2017, biggest sports festival in North India.
2. Served as Vice- Captain of Athletics Team at IIT(BHU) and lead the team at Inter-IIT Meet at IIT Madras.
3. Participated in Inter-IIT tournament at IIT-Kanpur and IIT- Madras for Javelin Event.